RED SEAL EXAM PREP

306A Plumber

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Grouped by this site's own topic areas
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Mock Exam — answer all questions, then check the key

Q1. **What is the purpose of an air admittance valve (AAV)?**

A) To admit outside air into a septic system

B) To supply air to the combustion chamber of a water heater

C) To bleed air from pressurized water supply lines

D) To admit air into the drain to stop siphonage

Q2. **A tubing cutter produces a better result than a hacksaw for copper pipe because:**

A) It is faster for large-diameter pipe

B) It creates a smooth flare automatically

C) It produces a square, undeformed cut

D) It does not require reaming

Q3. **What material is NOT approved for gas piping in residential construction?**

A) PVC plastic pipe, buried or exposed

B) Corrugated stainless steel tubing (CSST)

C) Copper tubing sized to the gas code tables

D) Threaded black steel pipe, Schedule 40

Q4. **A 306A plumber finds that a lavatory drain is running very slowly but is not completely blocked. Snaking the drain clears it, but it becomes slow again within 2 weeks. What does this pattern MOST likely indicate?**

A) The P-trap is partially blocked with hair — annual cleaning only solution

B) The drain is the wrong size for the fixture

C) The drain has a belly (sag) that traps solids

D) The trap seal is deeper than required, holding waste in the trap

Q5. **Why are S-traps not accepted as fixture traps in modern plumbing?**

A) S-traps self-siphon and lose their seal

B) They require too much head room below the fixture

C) They are too expensive to install and maintain

D) They are not compatible with modern ABS pipe

Q6. **What is the purpose of a P-trap access under a kitchen sink?**

A) To increase the volume of the trap seal

B) Aesthetic only — not functional

C) To allow the trap to be cleaned in place

D) To comply with commercial code only

Q7. **What is the purpose of a P-trap in plumbing?**

A) To reduce water hammer in the drain line

B) To filter debris out of the drain line

C) To hold a water seal that blocks sewer gas

D) To increase water pressure to the fixture

Q8. **What is the purpose of a drip leg (sediment trap) in a gas line?**

A) To reduce gas pressure to the appliance

B) To trap condensate and debris before the appliance

C) To serve as a shutoff valve location

D) To equalize pressure between high and low pressure gas

Q9. **What is the function of a fill valve (ballcock) in a toilet tank?**

A) To control the flush valve

B) To regulate water pressure to the toilet

C) To flush the toilet bowl

D) To refill the tank after a flush

Q10. **What is the purpose of the "flood level rim" reference in plumbing code?**

A) The fixture edge from which water would overflow

B) The top of a water closet tank

C) The maximum water level allowed in a river before flooding affects plumbing

D) The maximum fill level of a water heater

Q11. **What is a potable water system cross-connection?**

A) A potable water line that crosses from one building to another over a property line

B) Any tee fitting where a branch line is taken off the potable water distribution main

C) A connection between potable water and a contamination source

D) A point where the hot and cold supply lines are joined ahead of a mixing valve

Q12. **A 306A plumber wants to use a drainage arrangement that departs from the applicable acceptable solutions in Division B of the National Plumbing Code of Canada, and puts it forward as an alternative solution. Under that Code, who has to produce the supporting documentation, and what must it demonstrate?**

A) The plumber proposing it, showing it is adequate and safe in service

B) The manufacturer of the components, in a written product approval

C) The plumber proposing it, showing it performs as well as Division B

D) The authority having jurisdiction, in the written review it prepares

Q13. **What is a manifold distribution system in residential plumbing?**

A) A system where all fixtures connect to a single main shutoff

B) A home-run system with individual lines from a central manifold

C) A recirculation manifold for hot water

D) A system using a pressure manifold to equalize pressure to all floors

Q14. **Where must a main water supply shutoff valve be located?**

A) At the street property line

B) At the outdoor water meter location only

C) In the basement only

D) Inside the building at the service entry

Q15. **What is the standard leak test for gas piping after installation per CSA B149.1?**

A) Leak detection spray at fittings only

B) Visual inspection of joints only

C) Pressure test with air or nitrogen

D) Fill with water and check for leaks

Q16. **ABS pipe joints are made using:**

A) Compression fittings with rubber O-rings

B) ABS solvent cement with no separate primer

C) Threaded connections cut into the pipe wall

D) PVC purple primer followed by ABS solvent cement

Q17. **A pipe wrench should NOT be used on which type of pipe?**

A) Cast iron pipe

B) Black steel pipe

C) Galvanized steel pipe

D) Copper or plastic pipe

Q18. **What is the purpose of the authority having jurisdiction (AHJ) in plumbing inspections?**

A) To train plumbing apprentices

B) To certify plumbing products and materials

C) To inspect work and enforce the plumbing code

D) To sell plumbing permits to homeowners

Q19. **What is required when a plumber discovers unapproved (non-permitted) plumbing work during a renovation?**

A) Proceed with the renovation and do not report it

B) The plumber must refuse all work on the property

C) Inform the owner and recommend a permit and inspection

D) Nothing — historical non-permitted work is grandfathered

Q20. **What is the purpose of an overflow assembly in a bathtub?**

A) To connect the tub drain to the vent stack

B) To prevent overflow if the drain is closed

C) To allow draining the tub from both ends

D) To increase drain capacity for a filled tub

Q21. **What is the purpose of a temperature and pressure (T&P) relief valve on a water heater?**

A) To limit the hot water delivered to the fixtures to a safe tap temperature

B) To absorb thermal expansion in a closed system so the tank is not overstressed

C) To drop incoming cold water pressure down to the heater's rated working pressure

D) To release water automatically on excess pressure or temperature

Q22. **What is the correct procedure before opening a gas line for work?**

A) The line self-purges when opened

B) Shut off the gas, purge the line, and test

C) Open windows for ventilation only

D) Work quickly to minimize gas release

Q23. **Propane (LP gas) vapour is heavier than air. What is the key safety implication?**

A) Propane detectors must be mounted at ceiling level

B) Propane vents must exhaust upward through the roof

C) Propane gas cannot escape through floor drains

D) Propane accumulates in basements and low areas

Q24. **A 306A plumber is installing a 100mm (4") diameter cast iron drain stack in a 3-storey building. Where the stack changes from vertical to horizontal at its base, what fitting arrangement is acceptable for the transition to the building drain?**

A) A long-sweep 90° elbow or two 45° elbows may be used at the base

B) No 90° fittings of any kind — only two 45° elbows may be used

C) A short-pattern (short-radius) 90° elbow is permitted at the base of the stack

D) Any fitting is acceptable if the stack serves 3 storeys or less

Q25. **A stack in a 6-storey apartment building drains fixtures on every storey and does not serve as a wet vent, so the plumbing code requires a vent stack for it. Where must the lower end of that vent stack be connected?**

A) To a vertical section of the stack, immediately above the highest drainage connection to it

B) To a vertical section of the stack, at least 1 m above the floor level of the lowest storey

C) To the horizontal building drain, immediately downstream of the base of the stack served

D) To a vertical section of the stack, at or immediately below the lowest drainage connection

Q26. When threading black steel gas pipe, the correct pipe thread standard is:

A) NPT (National Pipe Taper)

B) BSP (British Standard Pipe)

C) NPS (National Pipe Straight)

D) Metric ISO threads

Q27. What type of valve is used to supply water to a toilet tank?

A) Ball valve (full bore)

B) Angle stop valve

C) Globe valve (in-line)

D) Gate valve (rising stem)

Q28. When using a pipe threading machine, the correct lubricant to use is:

A) Motor oil

B) Water

C) Threading (cutting) oil

D) No lubricant needed for power threading

Q29. What is the minimum slope (grade) for a 3-inch (75 mm) horizontal drain pipe?

A) No slope required for drain pipes

B) 2% (1/4 in per foot)

C) 4% (1/2 in per foot)

D) 1% (1/8 in per foot)

Q30. A hot water heater is producing discoloured (rusty) water but the cold water is clear. What should the plumber suspect FIRST and recommend?

A) Water softener malfunction — check salt levels

B) Hot water piping corrosion — replace all copper piping

C) The water main has a main break upstream — flush the system

D) The anode rod has depleted — replace the sacrificial anode

Q31. A new potable water distribution system has passed its pressure test. What must be done before the building is occupied?

A) Clean, flush and sanitize the system before occupancy

B) Rely on the chlorine already in the municipal supply

C) Run every fixture until the water is clear and odourless

D) Add a scale inhibitor chemical to the potable lines

Q32. What are the two main fuel gases supplied to homes in Canada?

A) Hydrogen and methane

B) Methane and propane

C) Acetylene and propane

D) Butane and natural gas

Q33. **A plumber is testing a new DWV system and finds that a floor drain in the basement is gurgling every time an upstairs fixture drains. There is no evidence of blockage. What is the MOST likely cause?**

A) Inadequate venting at that branch is breaking the trap seal

B) The branch drain has been graded far more steeply than needed

C) The trap seal has evaporated because the drain is seldom used

D) The floor drain is wrongly connected to the storm system

Q34. **What is the purpose of a flaring tool in plumbing?**

A) To expand plastic pipe for push-fit connections

B) To cut copper tubing at precise angles

C) To form a conical bell on soft tubing ends

D) To ream out burrs inside cut pipe

Q35. **What is the purpose of a vacuum breaker on a hose bibb (outdoor faucet)?**

A) To prevent the hose from kinking under vacuum

B) To reduce water pressure at outdoor connections

C) To prevent backflow from a submerged garden hose

D) To prevent water hammer when the hose is disconnected

Q36. **What is the purpose of a gate valve vs. a ball valve in water supply?**

A) Ball valve: used for main shutoffs only; gate valve: for branch lines

B) They are interchangeable with no operational difference

C) Gate valve: fully open or closed only; ball valve: quarter-turn full bore

D) Gate valve for hot water; ball valve for cold water only

Q37. **What is the purpose of a wax ring when installing a water closet?**

A) To secure the toilet to the floor structurally

B) To absorb vibration from the toilet

C) To align the toilet with the drain

D) To seal the toilet horn to the closet flange

Q38. **What is a floor drain trap primer?**

A) A chemical treatment used to kill bacteria in floor drain traps

B) A device that automatically adds water to maintain the trap seal

C) A primer applied to the drain pipe before installing the trap

D) A backflow preventer fitted on the outlet of the floor drain

Q39. **A plumber is installing drain-waste-vent piping in a confined space (pump room, 4m x 3m x 2.5m) and must use a solvent cement (ABS or PVC cement). What safety precautions are REQUIRED before beginning work?**

A) Open one window and prop the access door, then carry on with the cementing

B) An N95 dust mask, safety glasses, and nitrile gloves before opening the cement

C) Work in 15-minute intervals and step outside for fresh air between each interval

D) Atmospheric testing, forced-air ventilation, and continuous monitoring

Q40. **What is the maximum length of a trap arm (distance from trap weir to vent)?**

A) Varies by pipe diameter — set by NPC venting tables

B) Unlimited as long as the required 1 in 50 slope is maintained throughout

C) Fixed at 600 mm (24 in) from the trap weir for every pipe size

D) One foot of developed length for each inch of trap arm diameter

Q41. **A 306A plumber is laying out the fixture drainage for a new building. The plumbing code requires that fixtures be protected by a separate trap, and then names the cases in which one trap is permitted to protect more than one fixture. Which of these is one of those cases?**

A) Two fixtures whose traps are within 1.2 m of each other

B) A water closet and the lavatory standing beside it

C) Two lavatories back to back in adjoining bathrooms

D) Two similar single-compartment fixtures in one room

Q42. **When is a sanitary drainage system required to have an interceptor other than a grease trap?**

A) When the drainage velocity in the branch exceeds 1 metre per second

B) When the waste could damage or impair the drainage system

C) Only where the building stands in an industrial zoning district

D) In every building that has more than five plumbing fixtures

Q43. **What is the difference between high-pressure and low-pressure gas distribution?**

A) High pressure serves commercial buildings; low pressure serves homes

B) High pressure uses CSST only; low pressure uses black steel only

C) A single pressure is used throughout the whole distribution system

D) High pressure serves street mains; low pressure serves appliances

Q44. **What is the minimum hot water temperature at the water heater to prevent Legionella growth?**

A) 45°C (113°F)

B) 70°C (158°F)

C) 80°C (176°F)

D) 60°C (140°F)

Q45. **A new building has water hammer occurring every time the washing machine valve closes. The static water pressure is 480 kPa and no air chambers were installed. What is the MOST effective permanent fix?**

A) Reduce water pressure at the main to 400 kPa

B) Add a capped air chamber pipe stub at the washing machine valve

C) Install a water hammer arrestor at the washing machine supply

D) Install a pressure-reducing valve at the building entry

Q46. **What is a wet vent?**

A) A vent pipe that admits outside air to the building drain

B) A vent pipe fitted with a trap to seal off sewer gas

C) A single vent serving several fixtures on one branch

D) A soil-or-waste pipe that also vents other fixtures

Q47. **What is the difference between a sanitary sewer and a storm sewer?**

A) Sanitary sewer is for commercial buildings; storm sewer is for residential

B) They are the same in modern buildings

C) Sanitary sewer carries sewage; storm sewer carries surface water

D) Sanitary sewer is deeper; storm sewer is shallower

Q48. **How is the minimum size of a vent pipe serving a fixture determined?**

A) From the size of the trap served, using the code table

B) As one third of the drainage pipe that the vent serves

C) As one half of the drainage pipe that the vent serves

D) The same size as the drainage pipe that the vent serves

Q49. **What is the normal cold water supply pressure in a residential building?**

A) 70–100 kPa (10–15 psi)

B) Above 1000 kPa (145 psi)

C) 700–1000 kPa (100–145 psi)

D) 200–550 kPa (30–80 psi)

Q50. **A 306A plumber is direct-connecting the discharge line of a residential dishwasher that has its own drainage pump into the fixture outlet pipe of the adjacent kitchen sink. What does the plumbing code require of that discharge line?**

A) Connect straight into the food waste disposal unit with no rise in the hose

B) Install a spring check valve in the drain hose close to the dishwasher

C) Connect the drain hose to the sink tailpiece, running it flat under the cabinet

D) Rise as high as possible under the counter, then to the sink trap inlet

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 130-question 306A bank; the last page shows what the complete printable bank contains.

Q1. Answer: D

An air admittance valve opens to admit air when negative pressure develops in the drain, preventing trap siphonage without running a vent pipe from that fixture up through the roof. It closes under positive pressure so sewer gas cannot escape through it.

Q2. Answer: C

A tubing cutter rolls a cutting wheel around the pipe, producing a square cut that does not distort the pipe out of round — unlike the ragged, out-of-square edge a hacksaw leaves. It does roll a small burr into the bore, so the inside edge must still be reamed before soldering.

Q3. Answer: A

PVC is not a fuel gas piping material. The gas installation code's materials section recognizes steel pipe, copper tube, steel tube, polyethylene, corrugated stainless steel tubing, corrugated metal connectors and hose connectors, and it requires that all plastic gas pipe, tube and fittings be polyethylene complying with CSA B137.4. A material the code does not name may be used only where it conforms to a nationally recognized standard or to a test report of a nationally recognized certification organization, and PVC has no such standing for fuel gas. Keep the two plastics apart: there is a plastic gas pipe in the trench, but it is polyethylene, and polyethylene is approved only for outdoor underground work - distribution mains, house service lines, and on-premise lines past the meter. Steel pipe is the common above-ground material, threaded, conforming to ASTM A53/A53M or A106 in Schedule 40, Schedule 80 or standard weight, and hot dipped zinc coated pipe and fittings may be used. Copper is on the list too, and the gas code does not ban it for propane: the code covers natural gas and propane together, and Type G copper tube is made to ASTM B837, 'Standard Specification for Seamless Copper Tube for Natural Gas and Liquified Petroleum (LP) Gas Fuel Distribution Systems'. Copper for gas is conditional - Types G, K and L above ground provided they do not exceed 1-3/8 in OD, and soft annealed Type K or externally coated tube underground - so a copper question is settled against the gas code in force and the authority having jurisdiction, not by a blanket rule either way.

Q4. Answer: C

A recurring slow drain that clears with snaking but returns quickly points to a structural problem needing a structural fix. A belly — a low sag in the horizontal drain caused by settlement or missing support — traps solids that accumulate until the pipe is slow again. Snaking removes the accumulation but not the belly, so the clock simply restarts. Camera inspection confirms the low spot; re-grading or replacing that section is the permanent fix. A deeper trap seal is not a cause of anything here — the trap is full of water either way, and a deep-seal trap is a remedy for evaporation, not a source of blockage.

Q5. Answer: A

An S-trap has both an upward and a downward dip, so its outlet turns straight down. The discharging water carries on down that second curve and pulls the water seal with it — the trap siphons itself dry and stops blocking sewer gas. The code reaches the same result by limiting the total fall of a trap arm to not more than the arm's inside diameter, which an S-trap cannot meet.

Q6. Answer: C

Kitchen sink P-traps (often slip-joint type under sinks) include a cleanout plug or can be disassembled to allow removal of grease blockages and retrieval of dropped items without cutting pipes or removing the entire drain system.

Q7. Answer: C

The P-trap retains a water seal, a trap seal depth of not less than 38 mm, that blocks sewer gas including hydrogen sulphide and methane from entering the living space through the drain.

Q8. Answer: B

A drip leg is a vertical pipe extension below the horizontal run, installed just before the gas appliance. It collects condensate, debris, and scale from the gas supply before they reach the appliance regulator or valve.

Q9. Answer: D

The fill valve (ballcock) refills the toilet tank and bowl to the correct water levels after a flush, then shuts off. It does not flush the toilet or control the flush: the flush valve, with its flapper, controls the release of water from the tank to the bowl when the lever is pressed. Nor does it regulate the supply pressure - where the static pressure at a fixture may exceed 550 kPa, the plumbing code calls for a pressure-reducing valve to limit it. A fill valve that does not shut off shows up as a running toilet: the water level keeps rising until it drains into the overflow pipe, and Fluidmaster's fix is to flush debris out of the fill valve and to replace the valve if that does not cure it. Do not confuse that with a toilet that refills on its own every few minutes or hours, which people call ghost flushing: Fluidmaster explains that refill sound as the toilet losing water - internally, from the tank into the bowl past a leaking flapper or flush valve, or through a refill tube pushed too far into the overflow pipe - or externally. The distinction to keep straight is fill valve (refills the tank) versus flush valve (empties it into the bowl).

Q10. Answer: A

The flood level rim (FLR) is the top edge of a fixture from which water would overflow (top of a sink bowl, edge of a bathtub). Code uses FLR to set minimum fixture placement height and supply/trap requirements.

Q11. Answer: C

A cross-connection is any actual or potential physical connection between the potable water supply and a non-potable source of contamination (irrigation systems, chemical tanks, sewer, etc.). Cross-connections must have appropriate backflow prevention.

Q12. Answer: C

The burden sits with whoever asks for the alternative solution: the plumber proposing it must document that it performs at least as well as the Division B acceptable solutions it departs from. The National Plumbing Code of Canada 2020 gives two routes to compliance and no others - complying with the applicable acceptable solutions in Division B, or using alternative solutions that "will achieve at least the minimum level of performance required by Division B in the areas defined by the objectives and functional statements attributed to the applicable acceptable solutions". Division C then puts the paperwork where the proposal came from: documentation "shall be provided by the person requesting the use of an alternative solution" to demonstrate that it complies, and it must include a Code analysis of the analytical methods and rationales used to determine that the proposal reaches that level of performance, plus the qualifications, experience and background of whoever takes responsibility for the design. Wrong answers: arguing that the arrangement is adequate and safe in service is the commonest error, and the Code's own explanatory note shuts it down - the test is not "well enough" but "as well as" a design that would satisfy the applicable acceptable solutions. The authority having jurisdiction is where that documentation is submitted and where it is retained, not where it is written; the Code never asks the regulator to make the proponent's case. A manufacturer's product approval speaks for a component's own performance and says nothing about whether an arrangement that departs from Division B reaches the performance Division B sets.

Q13. Answer: B

A manifold (home-run) system uses a central manifold with individual PEX supply lines running directly to each fixture, eliminating branch tees. Provides individual shutoffs and reduces pressure drops.

Q14. Answer: D

NPC requires an accessible shut-off valve on the water service pipe, located inside the building as close as possible to where the water service enters, so the water to the entire building can be shut off in an emergency.

Q15. Answer: C

Gas piping is proved by pressurizing it with air or an inert gas such as nitrogen or carbon dioxide. The fuel gas itself is never used as the test medium, and oxygen is never used because it can ignite on contact with oil. Test pressure and minimum hold time are read from the pressure test table in CSA B149.1, which sets them by the system's working pressure, size and length rather than by any fixed multiplier: a 2 psi system 50 m long is held at 15 psi for at least 15 minutes, while the same system 70 m long must hold for 60 minutes. This test is done before any appliance is connected, with components rated below the test pressure, such as line pressure regulators, isolated or removed. Soap solution is used afterward to locate a leak the pressure drop has already revealed.

Q16. Answer: B

ABS drainage pipe is joined with a one-step ABS solvent cement. The cement itself softens both surfaces and they fuse as it cures, so no separate primer step is used. PVC is the material that needs the two-step treatment: primer first to soften the surface, then PVC cement. Carrying the purple-primer habit over from PVC does not improve an ABS joint and is not the ABS procedure. ABS is a solvent-weld material, so a DWV run is not made up with compression fittings, and plastic DWV pipe wall is not designed to carry cut threads.

Q17. Answer: D

Pipe wrenches are designed for steel pipe. Using them on copper or plastic pipe crushes the softer material, causing leaks and damage.

Q18. Answer: C

The AHJ (typically a municipal building department or provincial authority) enforces the adopted plumbing code by reviewing permit applications and plans, issuing permits, inspecting installations, and issuing compliance certificates.

Q19. Answer: C

Discovering unapproved work creates liability concerns. The plumber should document the finding, inform the owner, and recommend obtaining retroactive permits and inspections for the unapproved work to protect all parties.

Q20. Answer: B

The overflow assembly provides an emergency drain opening at the upper end of the tub. If the drain is blocked or closed and the tub fills, water enters the overflow, preventing spillage onto the floor.

Q21. Answer: D

The T&P relief valve carries two sensing elements in one body. The temperature element sits in the top of the tank and opens to keep the water from exceeding 99 C; the pressure element opens when the pressure in the tank reaches the tank's rated working pressure. Either condition on its own can rupture a heater, so the valve discharges water rather than let the vessel fail. Its discharge pipe must be at least the size of the valve outlet, rigid, sloped downward, unthreaded at the outlet, and terminated by an indirect connection above a floor drain or other safe location, with no shut-off valve anywhere on the line.

Q22. Answer: B

Before opening any gas line: shut off at the meter or upstream valve, purge the line of remaining gas, and use a calibrated gas detector to verify the atmosphere is below 10% LEL before any ignition sources.

Q23. Answer: D

Propane (SG=1.52) is heavier than air, so it settles in low-lying areas (basements, pits, trenches). Unlike natural gas (SG=0.55) which disperses upward, propane accumulates and can reach explosive concentrations.

Q24. Answer: A

Vertical-to-horizontal at a stack base: a long-sweep 90° elbow or two 45° elbows is acceptable — a short-pattern 90° is not. The high-velocity vertical flow at the base creates significant turbulence. A long-sweep (long-radius) 90° elbow, with a centreline radius at least equal to the pipe size, or two 45° elbows provides a gradual direction change that reduces turbulence and back-pressure. Short-pattern 90° elbows are restricted to horizontal-to-vertical direction changes, where flow accelerates rather than crashes into the outside of the turn.

Q25. Answer: D

A vent stack is required where a stack drains fixtures from more than 4 storeys containing plumbing fixtures, unless that stack serves as a wet vent, and the Code fixes where the pipe starts: it is connected to a vertical section of the stack at or immediately below the lowest sanitary drainage pipe connected to that stack. The definition of the pipe says the same thing from the other end - a vent stack is connected at its lower end to the stack at or below the lowest sanitary drainage pipe connection, and at its upper end either to a vent header or to outside air. Taking the lower end off the horizontal building drain past the foot of the stack does not satisfy that Article, which calls for a vertical section of the stack itself; the model codes used south of the border do allow a building-drain connection, which is why the idea travels. Connecting immediately above the highest drainage connection describes the other pipe entirely: the stack carried on past its highest drainage connection is the stack vent, and the Code requires the upper end of every stack to terminate in one. That "at or above the highest sanitary drainage pipe connection" wording does exist a few Articles later, but it governs the second offset relief vent on an offset stack, not the base of a vent stack. The 1 m dimension belongs to yoke vents: on a stack receiving discharge from fixtures on more than 11 storeys, the yoke vent connects to the vent stack at least 1 m above the floor level of the lowest storey of the section it relieves. Carrying that figure down to the vent stack's own connection puts the fitting a storey too high and leaves the base of the stack unrelieved.

Q26. Answer: A

Gas piping uses NPT (National Pipe Taper) threads, which create a pressure-tight seal as the tapered male and female threads engage. Must use yellow PTFE tape or pipe thread compound rated for gas.

Q27. Answer: B

Angle stop valves (also called angle valves, supply stop valves, or fixture shutoffs) are installed at each toilet, lavatory, and sink so that one fixture can be isolated for service or replacement without shutting off water to the whole building. Wrong answers: gate valves are used for main and riser shutoffs, not at fixtures, being bulky and slow to operate; a full-bore ball valve is an in-line isolating valve rather than a compact fixture stop, although the modern quarter-turn angle stop does use a ball internally; a globe valve throttles flow and is installed in line, not as a fixture stop. Angle stops come in quarter-turn and multi-turn types, and quarter-turn ball-type stops have largely replaced traditional multi-turn packing valves because they are less likely to seize open. Always install angle stops where they stay accessible for inspection and operation, under sinks and behind toilets, never buried in a wall.

Q28. Answer: C

Threading oil (cutting oil) lubricates and cools the die head, flushes the chips out of it, and lets the dies form a clean, full thread - which is what makes a tight, leak-free joint and what keeps the dies sharp. Wrong answers: motor oil is a hydrodynamic lubricant with no extreme-pressure sulfur or chlorine additives, so it will not stop the die teeth from welding and galling to the pipe, and it does not carry chips out of the die head - the threads come out torn. Water flash-rusts both the pipe and the dies and provides no boundary lubrication at all. Running a power threader dry is not an option either: the dies overheat within a few threads, the thread form is destroyed and the die set is ruined. Dark or black threading oil is the general-purpose product; clear or 'white' threading oil is used on stainless steel and on galvanized or potable-water work.

Q29. Answer: B

A 75 mm (3 in) horizontal sanitary drain is graded at 2% — 1 in 50, or 20 mm per metre, which is 1/4 in per foot (NPC 2020, Sentence 2.4.8.1.(1)). With too little grade the flow is too slow to carry solids along with it. The NPC sets a minimum, not a maximum: its Note A-2.4.8.1.(1) says steeper slopes and higher velocities help keep pipes clean by moving heavier solids that might tend to clog the pipes.

Q30. Answer: D

Rusty hot water with clear cold water indicates tank corrosion — the sacrificial anode rod in the hot water tank has failed. Magnesium or aluminum anode rods protect the glass-lined steel tank through galvanic action. Once depleted, the tank corrodes internally. Replacing the anode rod early extends tank life. If the tank interior is already significantly corroded, replacement of the whole tank may be more cost-effective.

Q31. Answer: A

Commissioning a potable system is more than a pressure test. The lines are first flushed to carry out construction debris, cutting oil and flux residue, then the system is sanitized and flushed again so that no sanitizing solution is left for an occupant to drink. Water running clear proves the flush worked, not that the piping is disinfected. The disinfectant residual in the municipal supply is intended to protect water already in the mains, not to disinfect newly installed pipe. Inhibitor chemicals belong in closed heating loops and must never be introduced into potable lines.

Q32. Answer: B

Natural gas distributed through municipal pipelines is essentially methane (CH_4). Propane (LP gas/LPG) is delivered as a liquid to a cylinder or tank and serves homes beyond the reach of the gas mains. The two are separate fuels, not two grades of one — Canada's installation code is titled for both natural gas and propane, and an appliance is orificed and regulated for one or the other.

Q33. Answer: A

A trap that gurgles with no blockage is telling you air is being driven through the seal, and that is a venting deficiency at that branch. A discharge from an upper floor sends a slug of water down the stack and a pressure swing with it. Above the falling slug the pressure goes negative and can siphon trap seals on the upper branches; ahead of the slug the air is compressed, so at the base of the stack the pressure goes positive and escapes through the nearest weak point - very often a basement floor drain trap, which gurgles and bubbles as air blows out through the seal. Either half of that pressure swing points to the same cure: vent the branch properly, with relief venting at the base of the stack where back pressure is the mechanism. An air admittance valve is not a substitute here - it only admits air, so it cannot relieve positive pressure, and the code permits it only in the limited situations it lists, such as island counters and fixtures at risk of frost closure.

Q34. Answer: C

A flaring tool expands the end of soft copper or aluminum tubing into a 45-degree conical bell that seats against the cone of the flare fitting body, and the flare nut behind it draws the two together into a metal-to-metal seal - no solder and no flame. Wrong answers: taking the internal burr off a cut tube end is the reamer's or deburring tool's job and is done before flaring, not by the flaring tool; plastic pipe is not flared at all - PEX-a is joined with a cold-expansion tool and a reinforcing ring, and push-fit fittings need no expansion whatever, only a square, deburred, depth-marked tube end; and cutting tube squarely is a tube cutter's job, since a flaring tool has no cutting edge. The flare must seat fully and evenly - a cracked, oval or partial flare will leak. The 45° SAE flare is the plumbing, HVAC and LP-gas standard, while the 37° AN/JIC flare belongs to hydraulic and fuel systems.

Q35. Answer: C

A hose connection vacuum breaker prevents backflow contamination when a garden hose is lying submerged in contaminated water (pool chemicals, pesticides) and supply pressure drops, which could back-siphon the contamination into the potable supply.

Q36. Answer: C

Gate valves open/close via a rising stem and gate — designed for fully open or fully closed positions only (throttling causes erosion). Ball valves use a quarter-turn with a drilled ball — full bore, reliable, and suitable for all plumbing water service.

Q37. Answer: D

The wax ring creates a flexible, watertight seal between the toilet's horn (outlet) and the closet flange, preventing sewer gas and sewage leaks at the floor connection.

Q38. Answer: B

Floor drain traps can dry out (evaporate) if seldom used. A trap primer automatically supplies a small amount of water to maintain the seal, preventing sewer gas entry.

Q39. Answer: D

Solvent cement contains volatile organic compounds — flammable, harmful to breathe, and heavier than air. Entry into a confined space where they will be used calls for atmospheric testing before entry, forced-air ventilation, continuous monitoring while the work goes on, and a trained attendant outside. Acceptable atmospheric levels are oxygen of at least 19.5% and not more than 23% by volume, a flammable gas or vapour concentration less than 10% of its lower explosive limit for cold work such as solvent cementing, and exposure to contaminants below the applicable occupational exposure limit. Watch the upper oxygen figure: 23% is the Canadian limit, while 23.5% is the United States figure and does not apply here. A solvent-wet confined space can build vapour to explosive levels quickly, and because the vapour is denser than air it collects low down — exactly where a worker kneeling to make up a joint is breathing. An N95 filters particulate and does nothing about organic vapour, and neither a propped door nor stepping outside between intervals is a substitute for testing, ventilating and monitoring.

Q40. Answer: A

The maximum trap arm length (distance from trap weir to the vent connection) is specified in NPC tables based on pipe diameter. Exceeding this causes trap siphonage.

Q41. Answer: D

The rule is a separate trap for each fixture, and the exceptions are a short, closed list. Every fixture is still trap-protected either way; what an exception permits is one trap protecting more than one fixture. Two similar single-compartment fixtures located in the same room are on that list, alongside all the trays or compartments of a 2- or 3-compartment sink and a 2-compartment laundry tray. A separate sentence lets one trap serve a group of floor drains or shower drains, a group of washing machines or a group of laboratory sinks, provided those fixtures are in the same room and are not located where they can receive food or other organic matter. Read the two conditions in the wording as a pair, because each wrong answer here fails one of them. A water closet and the lavatory beside it are in one room but are not similar fixtures, so each takes its own trap. Two lavatories back to back are similar fixtures, but adjoining bathrooms are not one room, and a shared wall is not a shared room. The 1.2 m figure is real but belongs to a different rule: it caps the developed length of a fixture outlet pipe, and the code sets no proximity distance that entitles two fixtures to share a trap. Where one trap does serve two fixtures, it still has to be accessible for cleaning and vented like any other trap.

Q42. Answer: B

An interceptor is triggered by what the waste contains, not by the size, velocity or zoning of the installation. The general rule is that where a fixture or equipment discharges sewage or waste that may damage or impair the sanitary drainage system, or the functioning of a public or private sewage disposal system, provision must be made to treat that waste before it is discharged. The specific requirements outside grease are an oil interceptor where the discharge from a fixture may contain oil or gasoline - service garages, vehicle repair bays, fuelling stations - and a purpose-designed interceptor where a fixture discharges sand, grit or similar materials, as at a car wash or machine shop. Fixture count and drainage velocity trigger nothing.

Q43. Answer: D

Gas leaves the utility's supply mains at a pressure far above anything an appliance can accept, and it is stepped down in stages on the way to the burner. A service regulator at the meter, installed by the utility on a natural gas system, reduces the supply to the building's line pressure. Where that line pressure is still above what the appliance is rated for, a line pressure regulator brings it down to the 0.5 psig (14 in WC) maximum inlet pressure most residential and light commercial appliances accept. The appliance's own regulator then sets burner manifold pressure, roughly 3 to 4 in WC for natural gas and 10 to 11 in WC for propane. Pressure class does not decide the piping material, and commercial and residential buildings are fed from the same high-pressure mains.

Q44. Answer: D

Legionella grows vigorously in warm water, roughly 25°C to 45°C, and is killed by prolonged exposure above 60°C. That is why a storage water heater is kept at 60°C rather than at a comfortable delivery temperature, and why the code fixes the thermostat setting for an electric storage-type service water heater at 60°C. A thermostatic mixing valve then blends the stored water down so that what reaches the fixture cannot scald. Note the direction of each figure: 60°C is a floor in the tank, and the delivery figure is a ceiling at the fixture.

Q45. Answer: C

A factory-built water hammer arrestor is the permanent fix; an air chamber is not. A capped pipe stub waterlogs and stops working, and PDI-WH 201 warns that even a correctly sized air chamber only controls shock temporarily after it is installed. An arrestor uses a piston sealed against a permanently charged gas cushion, so it keeps absorbing the pressure wave each time a fast-acting solenoid valve slams shut. In Canada the device must conform to ANSI/ASSE 1010, "Water Hammer Arresters" — National Plumbing Code Sentence 2.2.10.15.(1) — and the duty to protect the system at all comes from Sentence 2.6.1.9.(1). A pressure-reducing valve is not the answer here: at 480 kPa the static pressure is below the 550 kPa trigger in Article 2.6.3.3., and lowering pressure does not absorb the surge a quick-closing valve creates. Size the arrestor by fixture-unit load from the PDI-WH 201 tables and install it on the branch serving that valve.

Q46. Answer: D

A wet vent is a soil-or-waste pipe that the code permits to serve as a vent as well. It carries the discharge of the fixtures upstream of it and at the same time vents fixtures connected to it, so one pipe does two jobs and no separate dry vent is run to those fixtures. The code allows the arrangement only where the conditions set out in its wet venting article are all met, and the pipe is sized from the hydraulic load draining into it using the venting load table rather than by rule of thumb. The wrong answers are each a different thing with its own name. A single vent serving several fixtures on one nominally horizontal branch is a circuit vent, which the code sets out in its own article separate from wet venting. A vent fitted with a trap is not a code arrangement at all: the crown-vented trap, the closest thing to it, is prohibited. And a pipe that admits outside air to the drainage system near the building trap is a fresh air inlet, another of the vent types the code lists alongside the wet vent.

Q47. Answer: C

Sanitary sewers carry domestic and commercial sewage and wastewater to treatment plants. Storm sewers collect rain and surface runoff, often discharging directly to waterways. In modern construction, they must be kept separate (separated sewer).

Q48. Answer: A

Vent size comes from the trap, not from a fraction of the drain. The code table of minimum permitted vent sizes is entered with the size of the trap served: a 1-1/4 in. trap takes a 1-1/4 in. vent, a 1-1/2 in. trap also takes 1-1/4 in., a 2 in., 3 in. or 4 in. trap takes a 1-1/2 in. vent, and a 5 in. or 6 in. trap takes a 2 in. vent. So a large trap does not produce a proportionally large vent, and no fixed fraction of the drain is used anywhere in the venting rules for fixture vents. The one fraction that does appear applies to stacks, not to fixture drains: a vent stack or stack vent is not less than one-half the size of the soil-or-waste stack at its base. A branch vent, stack vent, vent stack or vent header is also never smaller than the vent pipe it connects into.

Q49. Answer: D

Residential cold water systems are designed to work in the 200–550 kPa (30–80 psi) band measured at the fixtures. Below that band the upper-floor fixtures and any flush-valve fixtures lose flow, and a booster pump is the remedy. Above it, seats and washers wear quickly, water hammer gets worse and fixtures waste water, so a pressure-reducing valve is fitted where static pressure at a fixture would run too high. Judge the pressure where the fixture is, not at the meter: elevation and friction loss both come off the figure on the way up.

Q50. Answer: D

The pump discharge line must rise as high as possible to just under the counter before it drops to the connection. The adopted Code text reads that where a domestic dishwashing machine equipped with a drainage pump discharges through a direct connection into the fixture outlet pipe of an adjacent kitchen sink or disposal unit, the pump discharge line shall rise as high as possible to just under the counter and connect on the inlet side of the sink trap by means of a Y fitting, or to the disposal unit. That rise is the elevation break: waste standing or backing up in the sink outlet piping has to climb to counter height before it can reach the machine. A hose run flat under the cabinet gives no break at all. A check valve is a moving part that fouls on food solids, and the Code offers no check-valve substitute for the rise. Dropping straight into a food waste disposal unit without the rise leaves the same open path back to the machine - the disposal unit is a permitted connection point, but only at the bottom of the required rise. An air gap fitting mounted at the counter is a different arrangement that gives a physical break instead of relying on elevation; the Sentence quoted above is what governs the direct connection this question describes.

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